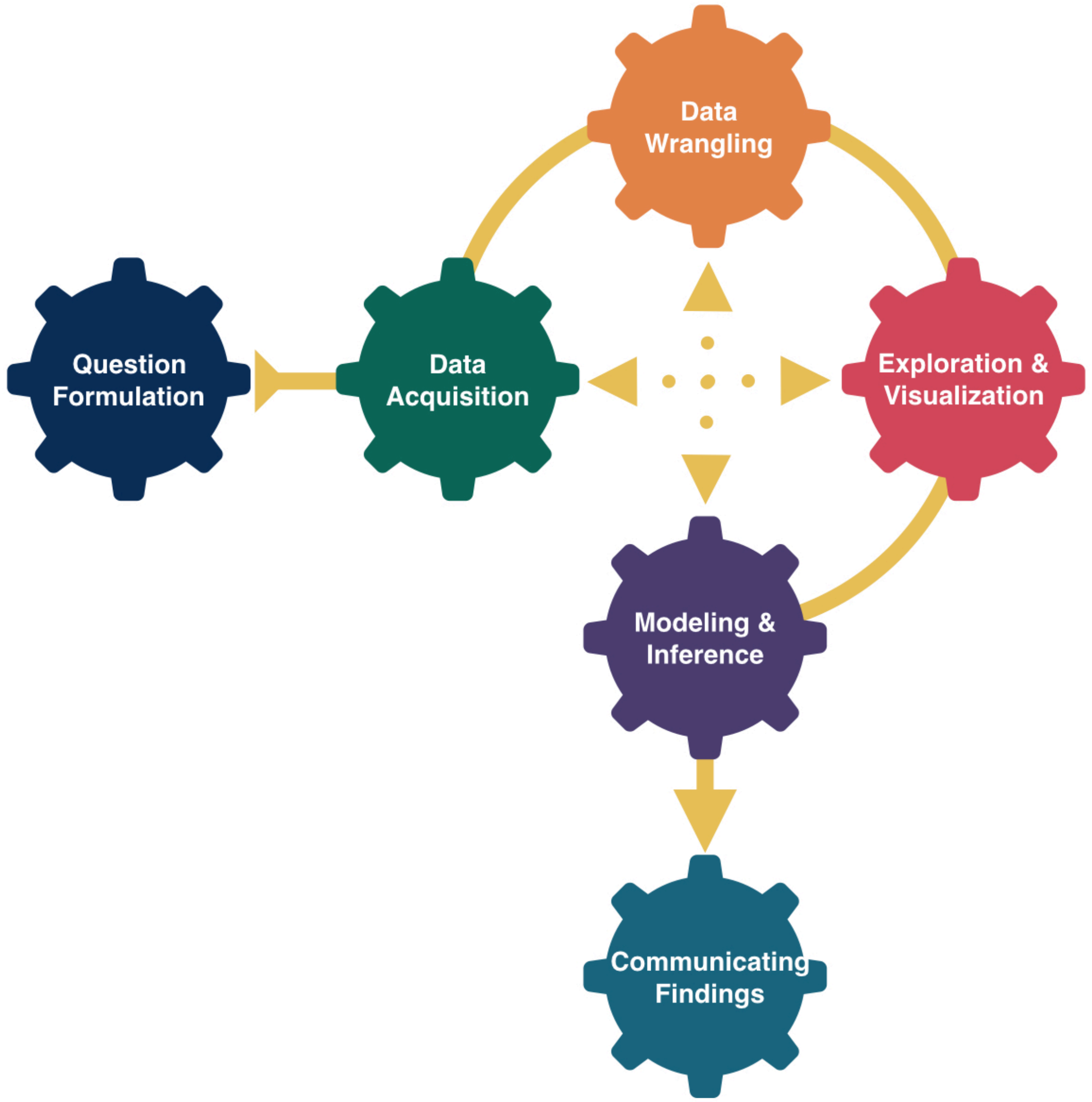

More Regression

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DATA 100

Week 7 | Spring 2026

Announcements

- Remember to come to office hours and post twice to Slack by Friday!

Goals for Today

- Handling categorical, explanatory variables with more than 2 categories
- Regression with polynomial explanatory variables

Linear Regression

Model Form:

$$y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

Linear regression is a flexible class of models that allow for:

- Both quantitative and categorical explanatory variables.
- Multiple explanatory variables.
- Curved relationships between the response variable and the explanatory variable.
- BUT the response variable is quantitative.

Example: Movies

Let's model a movie's critic rating using the audience rating and the movie's genre.

```r
library(tidyverse)
movies <- read_csv("https://www.lock5stat.com/datasets2e/HollywoodMovies.csv")

# Restrict our attention to dramas, horrors, and actions
movies2 <- movies %>%
  filter(Genre %in% c("Drama", "Horror", "Action")) %>%
  drop_na(Genre, AudienceScore, RottenTomatoes)
glimpse(movies2)
```

```
Rows: 313
Columns: 16
$ Movie            <chr> "Spider-Man 3", "Transformers", "Pirates of the Carib…
$ LeadStudio       <chr> "Sony", "Paramount", "Disney", "Warner Bros", "Warner…
$ RottenTomatoes   <dbl> 61, 57, 45, 60, 20, 79, 35, 28, 41, 71, 95, 42, 18, 2…
$ AudienceScore    <dbl> 54, 89, 74, 90, 68, 86, 55, 56, 81, 52, 84, 55, 70, 6…
$ Story            <chr> "Metamorphosis", "Monster Force", "Rescue", "Sacrific…
$ Genre            <chr> "Action", "Action", "Action", "Action", "Action", "Ac…
$ TheatersOpenWeek <dbl> 4252, 4011, 4362, 3103, 3778, 3408, 3959, 3619, 2911,…
$ OpeningWeekend   <dbl> 151.1, 70.5, 114.7, 70.9, 49.1, 33.4, 58.0, 45.3, 19.…
$ BOAvgOpenWeekend <dbl> 35540, 17577, 26302, 22844, 12996, 9791, 14663, 12541…
$ DomesticGross    <dbl> 336.53, 319.25, 309.42, 210.61, 140.13, 134.53, 131.9…
$ ForeignGross     <dbl> 554.34, 390.46, 654.00, 245.45, 117.90, 249.00, 157.1…
$ WorldGross       <dbl> 890.87, 709.71, 963.42, 456.07, 258.02, 383.53, 289.0…
$ Budget           <dbl> 258.0, 150.0, 300.0, 65.0, 140.0, 110.0, 130.0, 110.0…
```

Response variable:

Explanatory variables:

How should we encode a categorical variable with more than 2 categories?

Let’s start with what NOT to do.

Equal Slopes Model:

How should we encode a categorical variable with more than 2 categories?

What we should do instead.

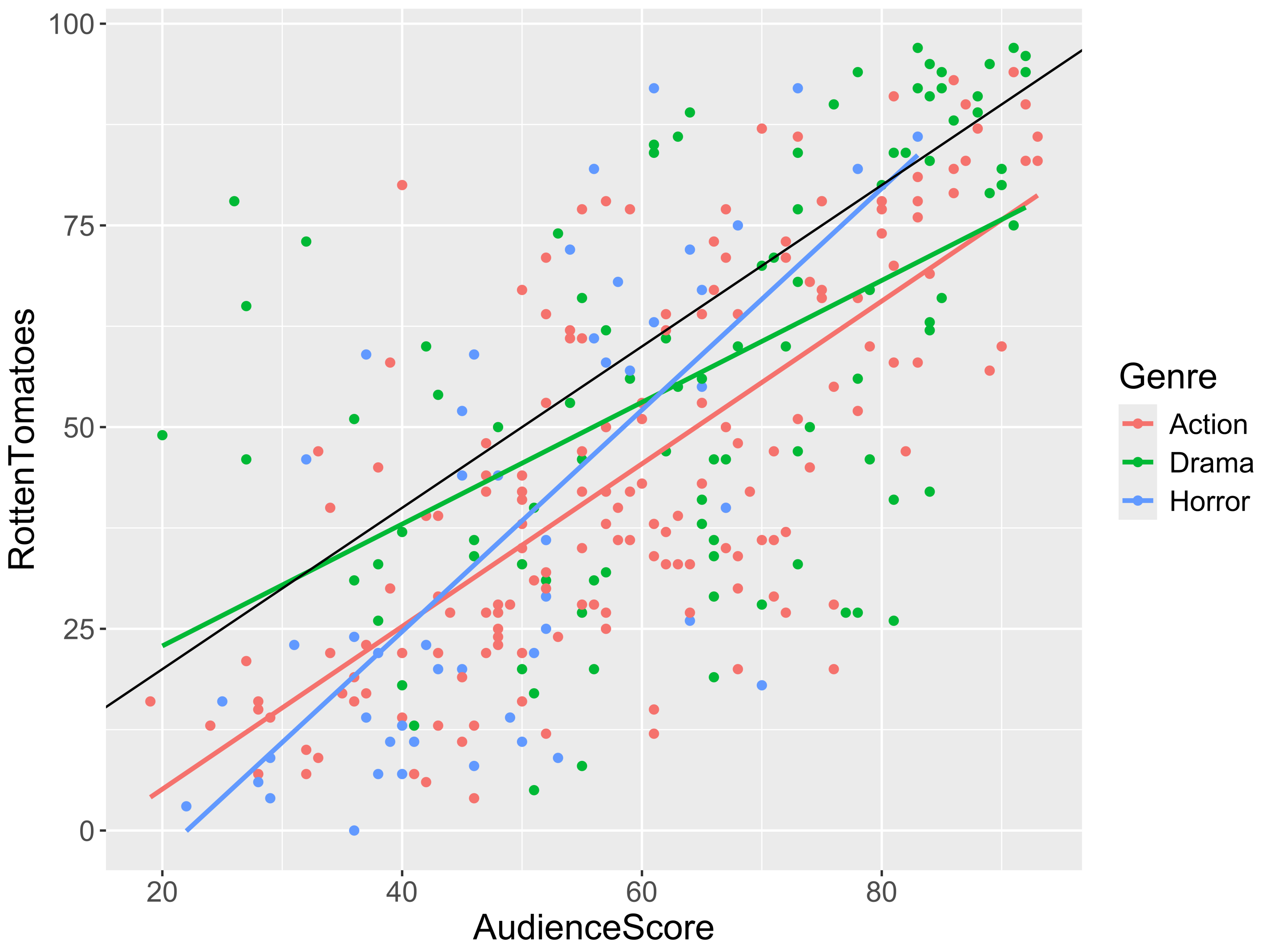
Equal Slopes Model:

How should we encode a categorical variable with more than 2 categories?

Different Slopes Model:

Exploring the Data

```r
ggplot(data = movies2,
       mapping = aes(x = AudienceScore,
                     y = RottenTomatoes,
                     color = Genre)) +
  geom_point() +
  stat_smooth(method = lm, se = FALSE) +
  geom_abline(slope = 1, intercept = 0)
```



- Trends?
- Should we include interaction terms in the model?

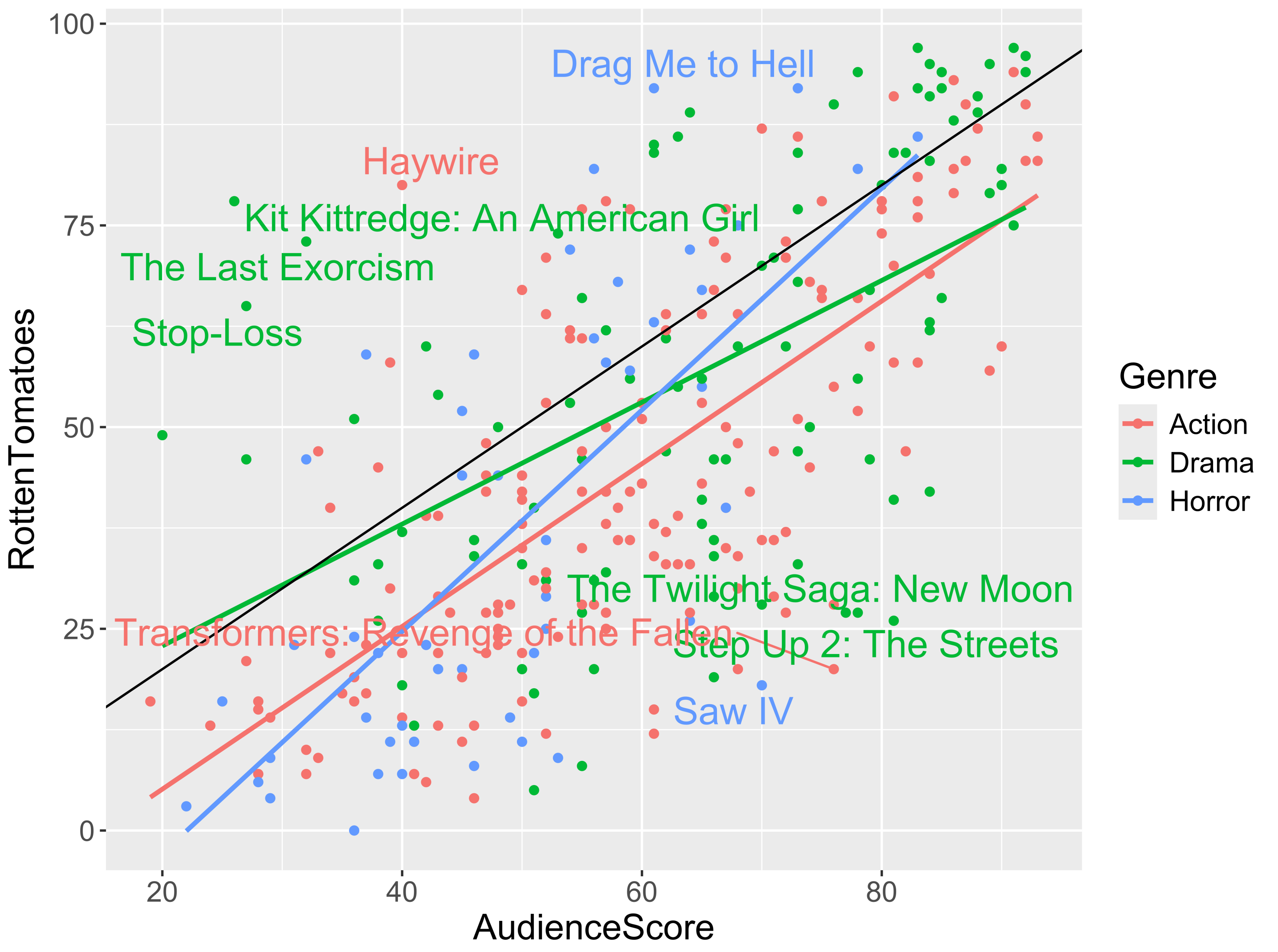
Side-bar: Identify Outliers on a Graph

```r
outliers <- movies2 %>%
  mutate(DiffScore = AudienceScore - RottenTomatoes) %>%
  filter(DiffScore > 50 | DiffScore < -30) %>%
  select(Movie, DiffScore, AudienceScore, RottenTomatoes, Genre)
outliers
```

```
# A tibble: 9 × 5
  Movie                              DiffScore AudienceScore RottenTomatoes Genre
  <chr>                                  <dbl>         <dbl>          <dbl> <chr>
1 Saw IV                                    52            70             18 Horr…
2 Step Up 2: The Streets                    55            81             26 Drama
3 Kit Kittredge: An American Girl          -52            26             78 Drama
4 Stop-Loss                                -38            27             65 Drama
5 Transformers: Revenge of the Fal…         56            76             20 Acti…
6 The Twilight Saga: New Moon               51            78             27 Drama
7 Drag Me to Hell                          -31            61             92 Horr…
8 The Last Exorcism                        -41            32             73 Drama
9 Haywire                                  -40            40             80 Acti…
```

Side-bar: Identify Outliers on a Graph

```
library(ggrepel)
ggplot(data = movies2,
       mapping = aes(x = AudienceScore,
                     y = RottenTomatoes,
                     color = Genre)) +
  geom_point() +
  stat_smooth(method = lm, se = FALSE) +
  geom_abline(slope = 1, intercept = 0) +
  geom_text_repel(data = outliers,
                  mapping = aes(label =
                                  Movie),
                  force = 10,
                  show.legend = FALSE,
                  size = 6)
```



Building the Model:

Full model form:

```
mod <- lm(RottenTomatoes ~ AudienceScore*Genre, data = movies2)

library(moderndive)
get_regression_table(mod)
```

```
# A tibble: 6 × 7
  term                      estimate std_error statistic p_value lower_ci upper_ci
  <chr>                        <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 intercept                  -15.0       5.27      -2.85   0.005  -25.4     -4.67
2 AudienceScore                1.01      0.085     11.8    0        0.84     1.18
3 Genre: Drama                22.8       8.94       2.55   0.011    5.23    40.4
4 Genre: Horror              -15.2      11.0       -1.39   0.165  -36.8      6.32
5 AudienceScore:GenreDra…    -0.253     0.136     -1.86   0.065   -0.522    0.015
6 AudienceScore:GenreHor…     0.365     0.206      1.77   0.078   -0.04     0.771
```

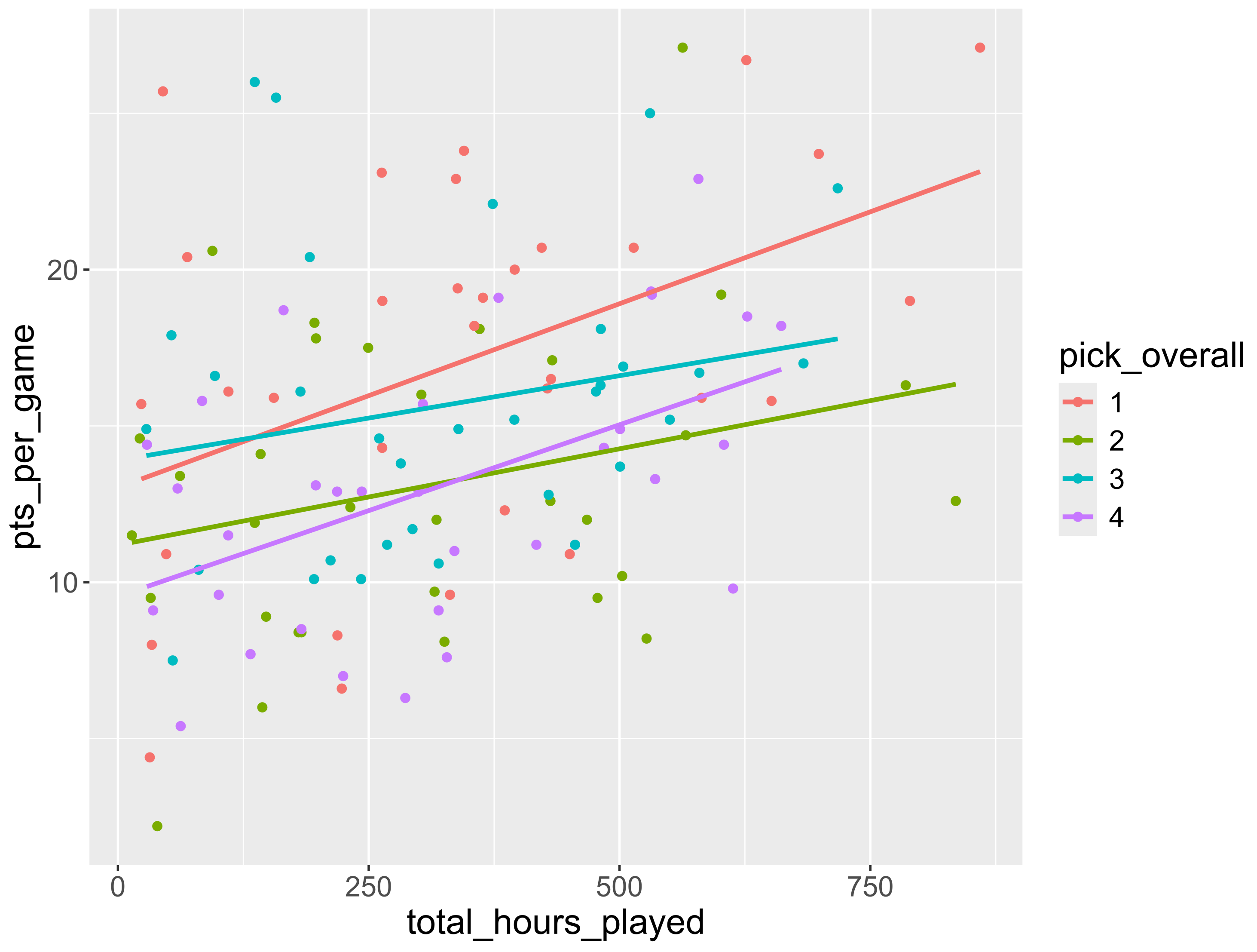
Estimated model for Dramas:

Let's Practice!

Here's data from 1990 - 2021 on the first four picks in the NBA draft. Let's explore how the relationship between time playing and points scored per game varies by a player's overall pick number.

```
nba_draft <- read_csv("https://data.scorenetwork.org/da
  filter(pick_overall <= 4) %>%
  mutate(pick_overall = as.factor(pick_overall),
         total_hours_played = total_mins_played/60)

ggplot(data = nba_draft, mapping = aes(
  x = total_hours_played, y = pts_per_game, color = pic
)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```



```r
mod1 <- lm(pts_per_game ~ total_hours_played + pick_overall, data = nba_draft)
get_regression_table(mod1)
```

```
# A tibble: 5 × 7
  term               estimate std_error statistic p_value lower_ci upper_ci
  <chr>                 <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 intercept            14.1       1.09      12.9    0        11.9     16.2
2 total_hours_played    0.009     0.002      4.30   0         0.005    0.013
3 pick_overall: 2      -3.68      1.18      -3.11   0.002    -6.02    -1.34
4 pick_overall: 3      -1.27      1.18      -1.08   0.285    -3.61     1.07
5 pick_overall: 4      -3.81      1.18      -3.23   0.002    -6.15    -1.47
```

```r
mod2 <- lm(pts_per_game ~ total_hours_played * pick_overall, data = nba_draft)
get_regression_table(mod2)
```

```
# A tibble: 8 × 7
  term                     estimate std_error statistic p_value lower_ci upper_ci
  <chr>                       <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 intercept                  13.0       1.55      8.40    0        9.96    16.1
2 total_hours_played          0.012     0.004     3.11    0.002    0.004    0.019
3 pick_overall: 2            -1.84      2.13     -0.863   0.39    -6.06     2.38
4 pick_overall: 3             0.87      2.31      0.377   0.707   -3.70     5.44
5 pick_overall: 4            -3.48      2.23     -1.56    0.121   -7.90     0.932
6 total_hours_played:pic…   -0.006     0.005    -1.03    0.303   -0.016    0.005
7 total_hours_played:pic…   -0.006     0.006    -1.08    0.283   -0.018    0.005
8 total_hours_played:pic…   -0.001     0.006    -0.135   0.893   -0.012    0.011
```

Practice

In Model 1, interpret $\hat{\beta}_1$.

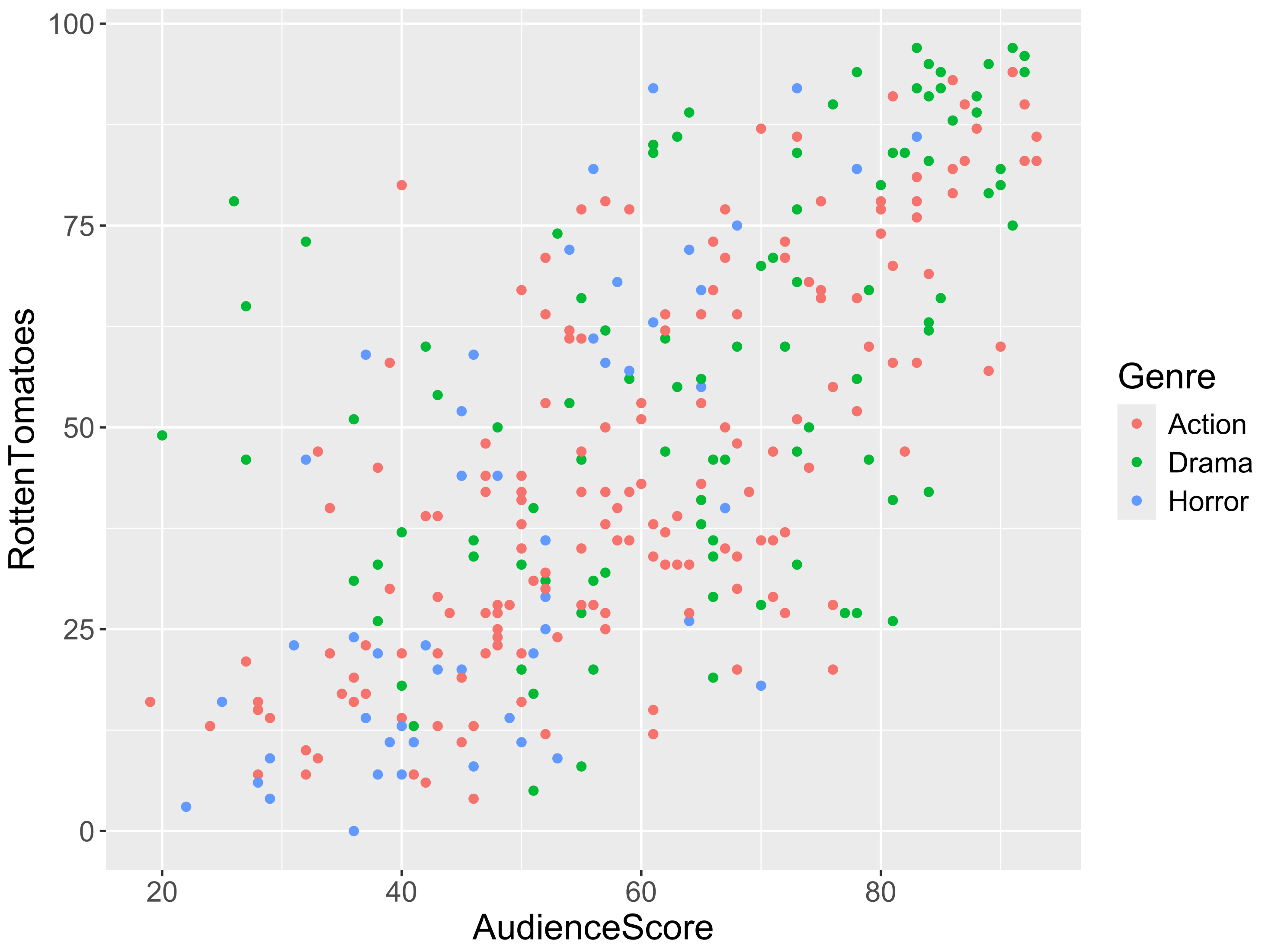
In Model 1, interpret $\hat{\beta}_2$.

For Model 2, determine the slope for the 1st picks, the slope for the 2nd picks, the slope for the 3rd picks, and the slope for the 4th picks.

Based on Model 2, which pick seems to make the best use of their play time (in terms of scoring points)?

Coming Back to Our Exploratory Data Analysis of Movies…

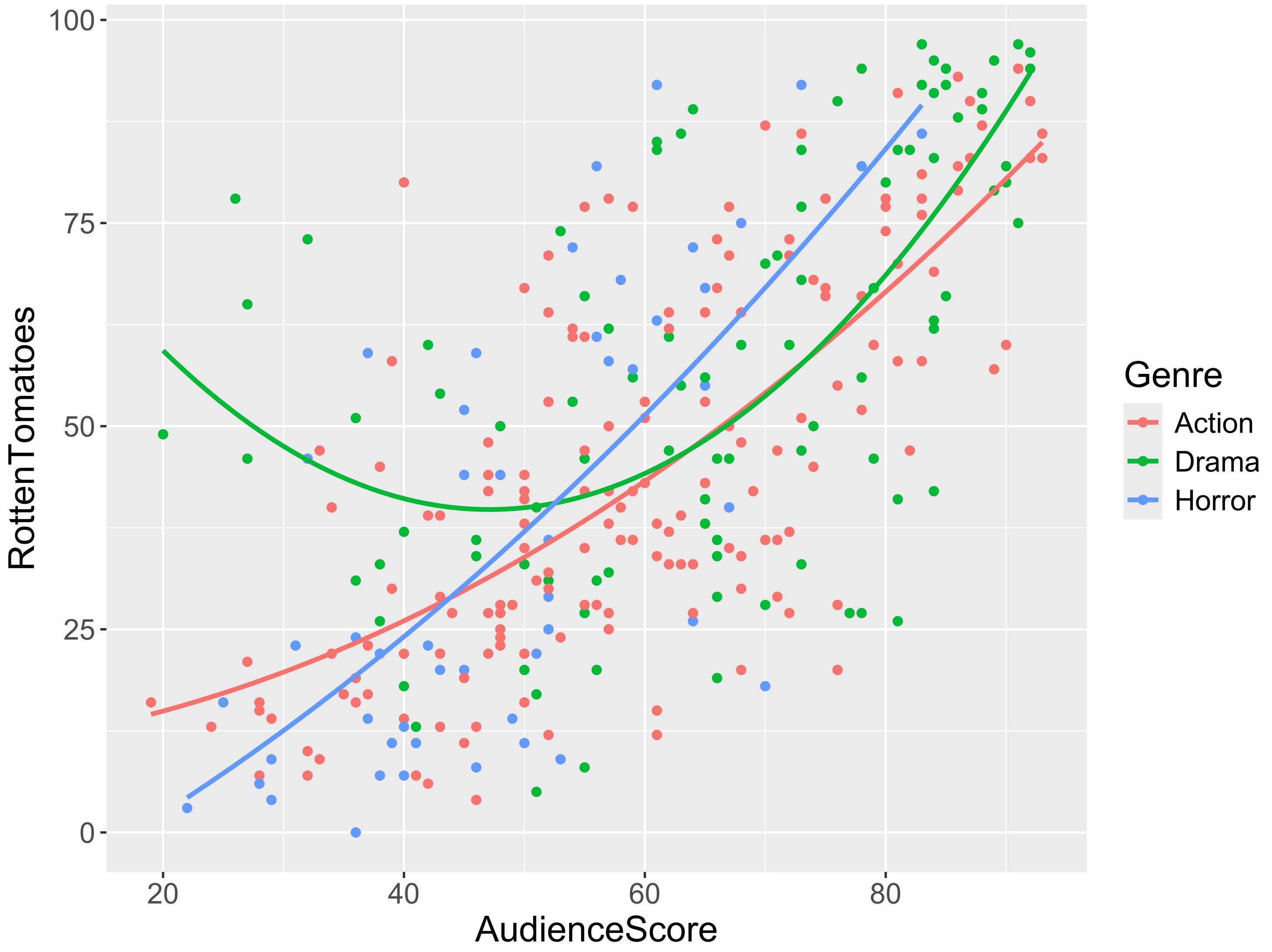
```r
library(ggrepel)
ggplot(data = movies2,
       mapping = aes(x = AudienceScore,
                     y = RottenTomatoes,
                     color = Genre)) +
  geom_point()
```



Evidence of curvature?

Adding a Curve to your Scatterplot

```r
ggplot(data = movies2,
       mapping = aes(x = AudienceScore,
                     y = RottenTomatoes,
                     color = Genre)) +
  geom_point() +
  stat_smooth(method = lm, se = FALSE,
        formula = y ~ poly(x, degree = 2))
```



Fitting the New Model

```r
mod2 <- lm(RottenTomatoes ~ poly(AudienceScore, degree = 2, raw = TRUE)*Genre,
           data = movies2)
get_regression_table(mod2, print = TRUE)
```

term	estimate	std_error	statistic	p_value	lower_ci	upper_ci
intercept	9.922	14.851	0.668	0.505	-19.301	39.145
poly(AudienceScore, degree = 2, raw = TRUE)1	0.098	0.515	0.191	0.849	-0.916	1.113
poly(AudienceScore, degree = 2, raw = TRUE)2	0.008	0.004	1.788	0.075	-0.001	0.016
Genre: Drama	88.923	24.538	3.624	0.000	40.638	137.208
Genre: Horror	-23.767	31.054	-0.765	0.445	-84.876	37.342
poly(AudienceScore, degree = 2, raw = TRUE)1:GenreDrama	-2.608	0.840	-3.107	0.002	-4.260	-0.956
poly(AudienceScore, degree = 2, raw = TRUE)2:GenreDrama	0.019	0.007	2.785	0.006	0.006	0.032
poly(AudienceScore, degree = 2, raw = TRUE)1:GenreHorror	0.574	1.223	0.469	0.639	-1.833	2.981
poly(AudienceScore, degree = 2, raw = TRUE)2:GenreHorror	-0.001	0.012	-0.061	0.951	-0.024	0.022

Linear Regression & Curved Relationships

Form of the Model:

$$y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + \epsilon$$

But why is it called linear regression if the model also handles for curved relationship??

Reminders:

- Remember to come to office hours and post twice to Slack by Friday!